

UNIT 07

ALCOHOLS, PHENOLS & ETHERS

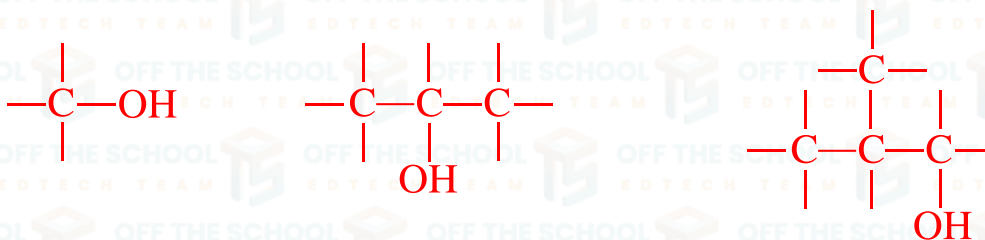


What Are Alcohols?

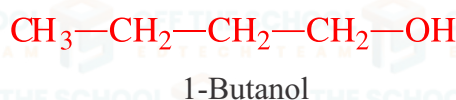
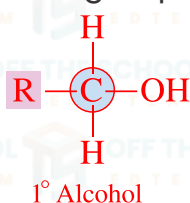
Definition: Alcohols are organic compounds where one or more hydrogen atoms in a hydrocarbon (a molecule made of hydrogen and carbon) have been replaced by a hydroxyl group ($-\text{OH}$). This group consists of an oxygen atom bonded to a hydrogen atom. The simplest example of an alcohol is methanol (CH_3OH), where the hydroxyl group is attached to a methyl group.

Classification of Alcohols on the basis of the nature of the carbon atom bonded to the $-\text{R}$ group:

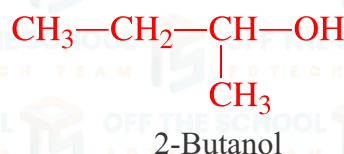
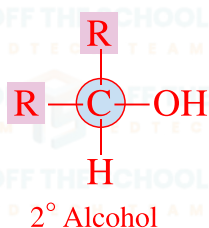
Alcohols are classified based on the number of carbon atoms connected to the carbon atom that carries the hydroxyl group:



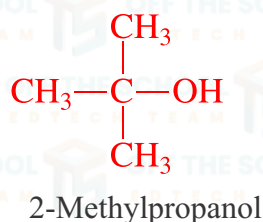
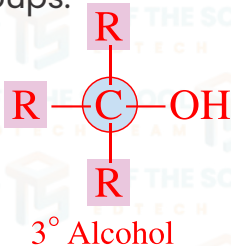
1. Primary (1°) Alcohols: In a primary alcohol, the carbon with the $-\text{OH}$ group is only attached to one other carbon atom. An example is ethanol ($\text{CH}_3\text{CH}_2\text{OH}$), where the $-\text{OH}$ group is on the end carbon.



2. Secondary (2°) Alcohols: In a secondary alcohol, the carbon with the $-\text{OH}$ group is attached to two other carbon atoms. An example is isopropanol ($(\text{CH}_3)_2\text{CHOH}$), where the $-\text{OH}$ group is on the middle carbon of a three-carbon chain.



3. Tertiary (3°) Alcohols: In a tertiary alcohol, the carbon with the $-\text{OH}$ group is attached to three other carbon atoms. An example is tert-butyl alcohol ($(\text{CH}_3)_3\text{COH}$), where the $-\text{OH}$ group is on a carbon atom connected to three methyl groups.



Classification of Alcohols Based on OH Groups

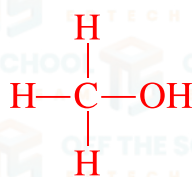
Alcohols can also be classified based on the number of hydroxyl groups they have:

1. Monohydric Alcohols: These alcohols have only one hydroxyl group attached to the carbon atom. An example is ethanol, which has the chemical formula $\text{CH}_3\text{CH}_2\text{OH}$.

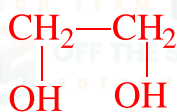


2. Dihydric Alcohols: These alcohols contain two hydroxyl groups. They are also known as glycols. An example is ethylene glycol (1,2-Ethandiol), which has the chemical formula $\text{HOCH}_2\text{CH}_2\text{OH}$ and is used in antifreeze.

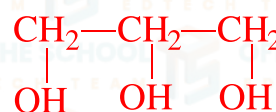
3. Trihydric Alcohols: These alcohols have three hydroxyl groups. They are also called triols. An example is glycerol (1,2,3-Propantriol), which has the chemical formula $\text{C}_3\text{H}_5(\text{OH})_3$ and is a common component of fats and oils.



Methanol



1,2-Ethanol



1,2,3-Propantriol

By understanding the number of hydroxyl groups, we can predict the properties and reactivity of different alcohols. The more hydroxyl groups an alcohol has, the more it can form hydrogen bonds, affecting its solubility and boiling point.

Boiling Point of Ether vs. Alcohol

Scientific Reason: Ether has a lower boiling point than alcohol because ethers generally do not form hydrogen bonds with each other. Alcohol molecules, on the other hand, can form strong hydrogen bonds due to the presence of the hydroxyl (OH) group, which results in higher boiling points.

Solubility of Alcohols in Water

Scientific Reason: Alcohols are soluble in water because the hydroxyl group in the alcohol molecule can form hydrogen bonds with water molecules. This interaction makes them miscible with water.

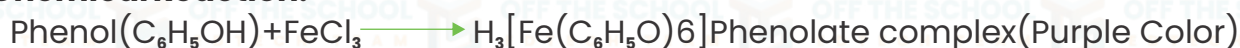
Ethanol vs. Ethyl Chloride at Room Temperature

Scientific Reason: Ethanol is a liquid at room temperature because of the hydrogen bonding between its molecules, which requires more energy to overcome. Ethyl chloride is a gas at room temperature because it is unable to form hydrogen bonds and has weaker van der Waals forces, requiring less energy to separate the molecules.

Laboratory Tests for Phenol

1. Iron (III) Chloride Test for Phenol:

Chemical Reaction:

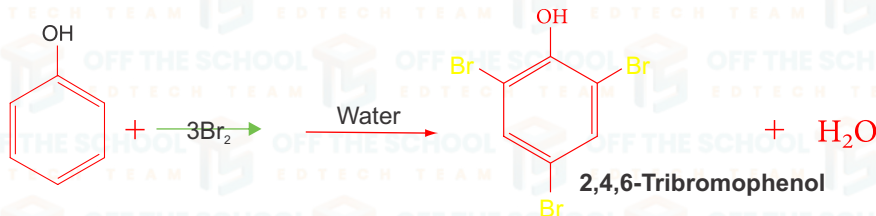


Procedure: When you add a few drops of iron (III) chloride to a solution of phenol, a purple complex is formed, which indicates the presence of phenol.

2. Bromine Water Test for Phenol

Chemical Reaction:





Procedure: If you add bromine water to a phenol solution, it reacts to form a white precipitate of 2,4,6-tribromophenol, confirming the presence of phenol.

Laboratory Tests for Alcohol:

1. Sodium Metal Test for Alcohol

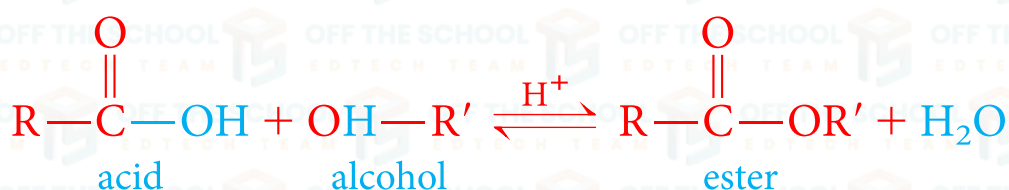
Chemical Reaction:



Procedure: When a small piece of sodium is added to alcohol, it reacts to form sodium alkoxide and hydrogen gas, which is observed as effervescence (bubbling).

2. Ester Test (Fischer Esterification) for Alcohol

Chemical Reaction:



Procedure: If you mix alcohol with a carboxylic acid (like acetic acid) and an acid catalyst (like sulfuric acid), then heat the mixture, it forms an ester. Esters often have a fruity smell, which is a positive result for this test.

Lucas Reagent

Definition: Lucas reagent is a solution of zinc chloride in concentrated hydrochloric acid.

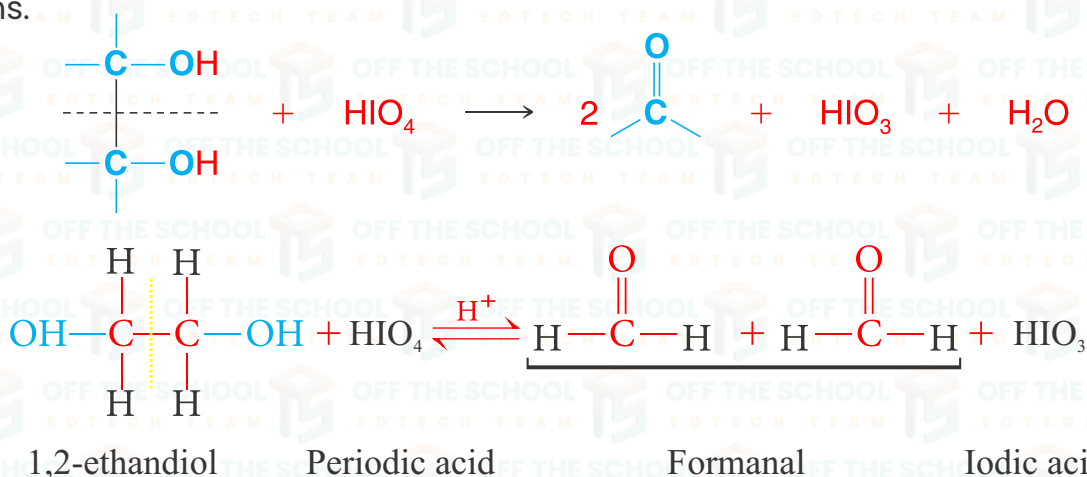
Use: It is used to distinguish between primary, secondary & tertiary alcohols.

Chemical Equation: For Primary alcohols:



Cleavage of 1,2-diols:

Cleavage of 1,2-diols, also known as vicinal diols, typically involves breaking the C-C bond between the two hydroxyl-bearing carbon atoms. This can be achieved through oxidative cleavage using reagents like periodic acid (HIO_4), which results in the formation of two carbonyl compounds, such as aldehydes, ketones, or a combination of both, depending on the substituents attached to the diol's carbon atoms.



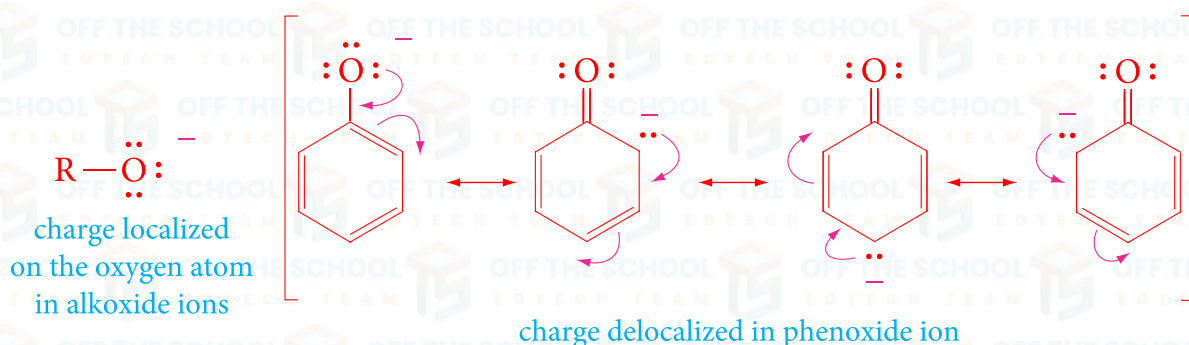
PHENOLS

Acidity of Phenols:

Phenols are stronger acids than alcohols:

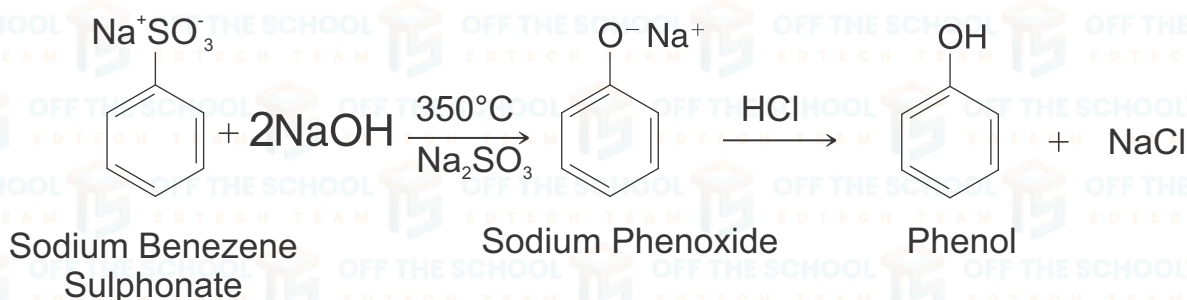
- Both alcohols and phenols have a hydroxyl group (-OH) that donates the proton, but phenols are stronger acids because the negative charge on the resulting ion is stabilized through resonance.
- In phenols, the negative charge on the oxygen can spread out onto the ring (ortho and para positions), making phenoxide ions more stable than alkoxide ions.

A simple chemical equation representing the dissociation of phenol into its anion is as follows:

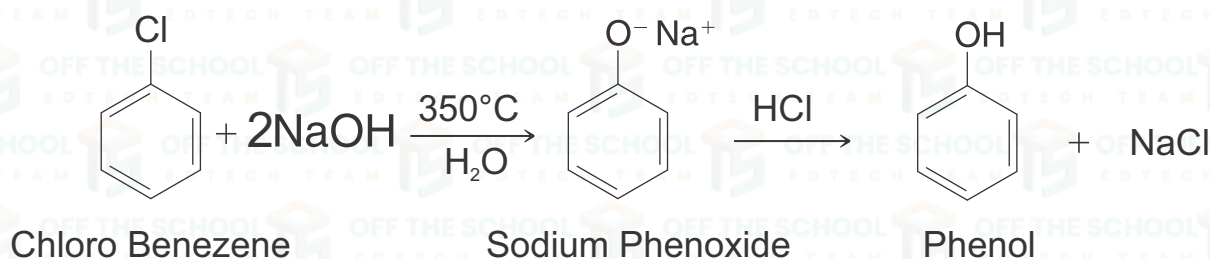


PREPARATION OF PHENOL

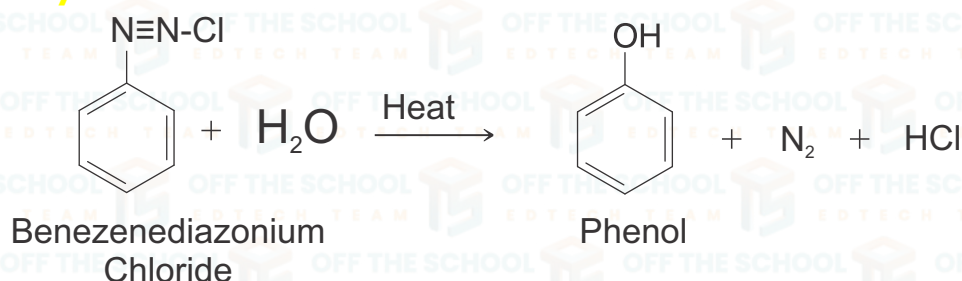
i) From Sodium Benzene Sulphonate:



ii) From Chlorobenzene (Dow's Process):



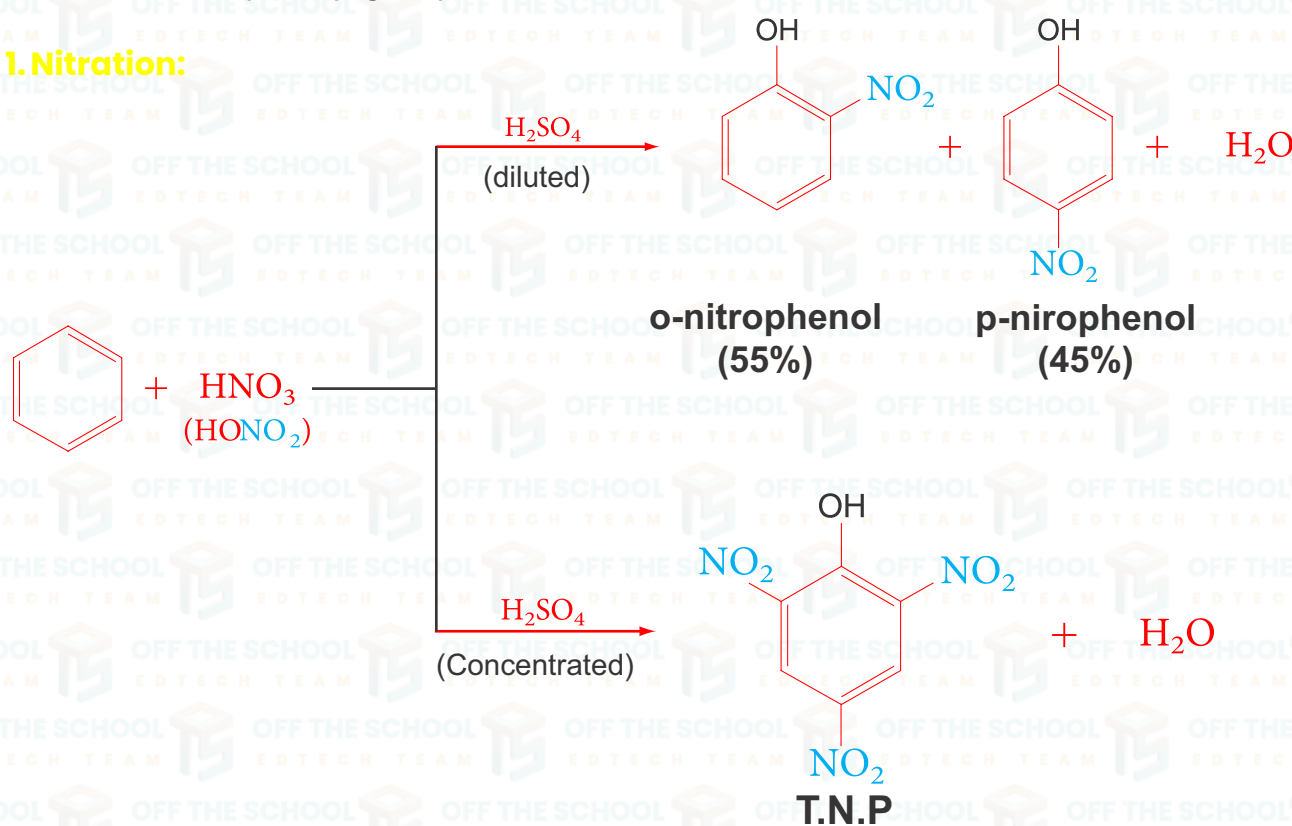
iii) From Hydrolysis of Diazonium salt:



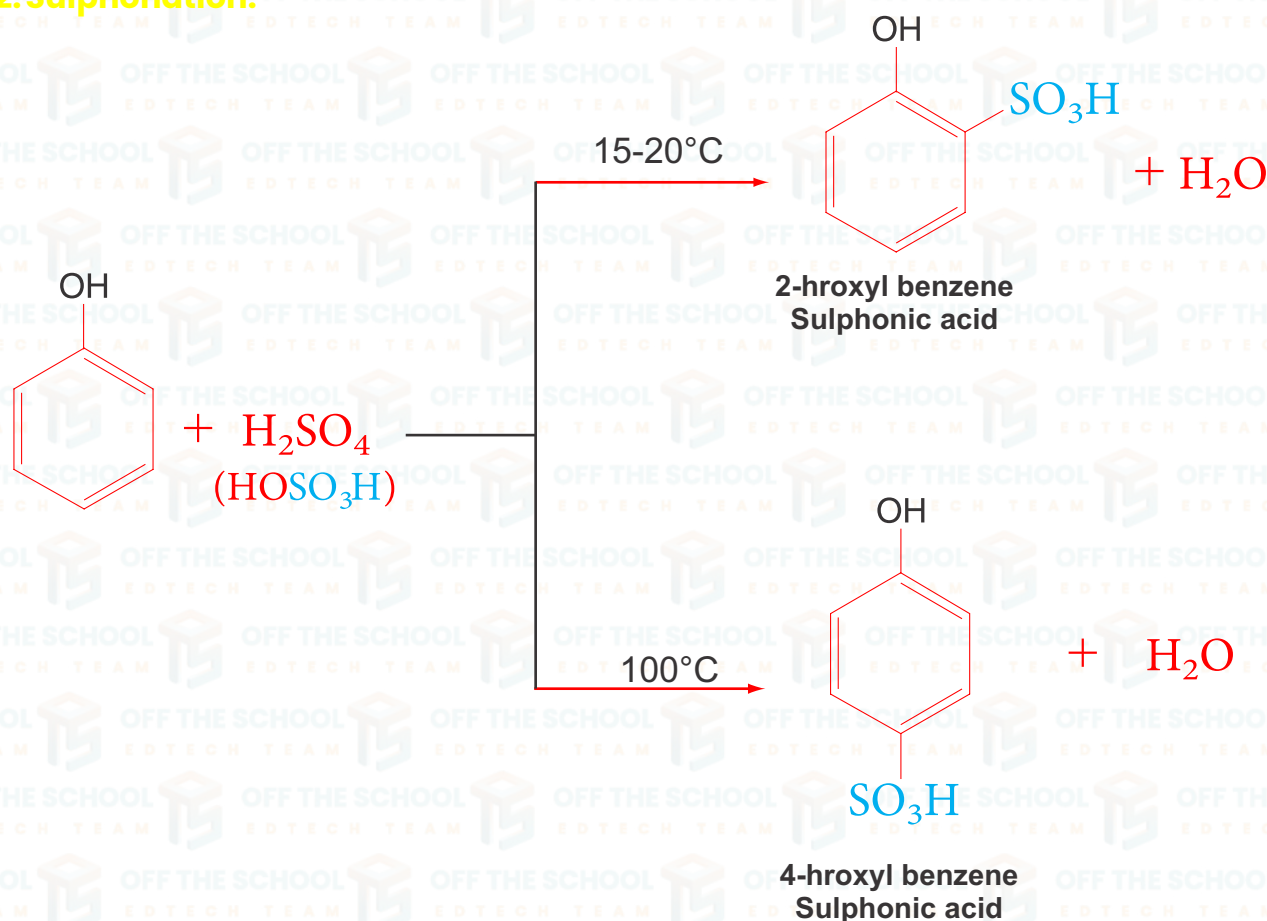
Electrophilic Aromatic substitutions in Phenols:

Electrophilic aromatic substitution in phenols involves the replacement of a hydrogen atom on the aromatic ring by an electrophile. The presence of the hydroxyl group in phenols makes the aromatic ring more reactive toward electrophilic substitution because it is an electron-donating group that increases the electron density on the ring, particularly at the ortho and para positions relative to the hydroxyl group.

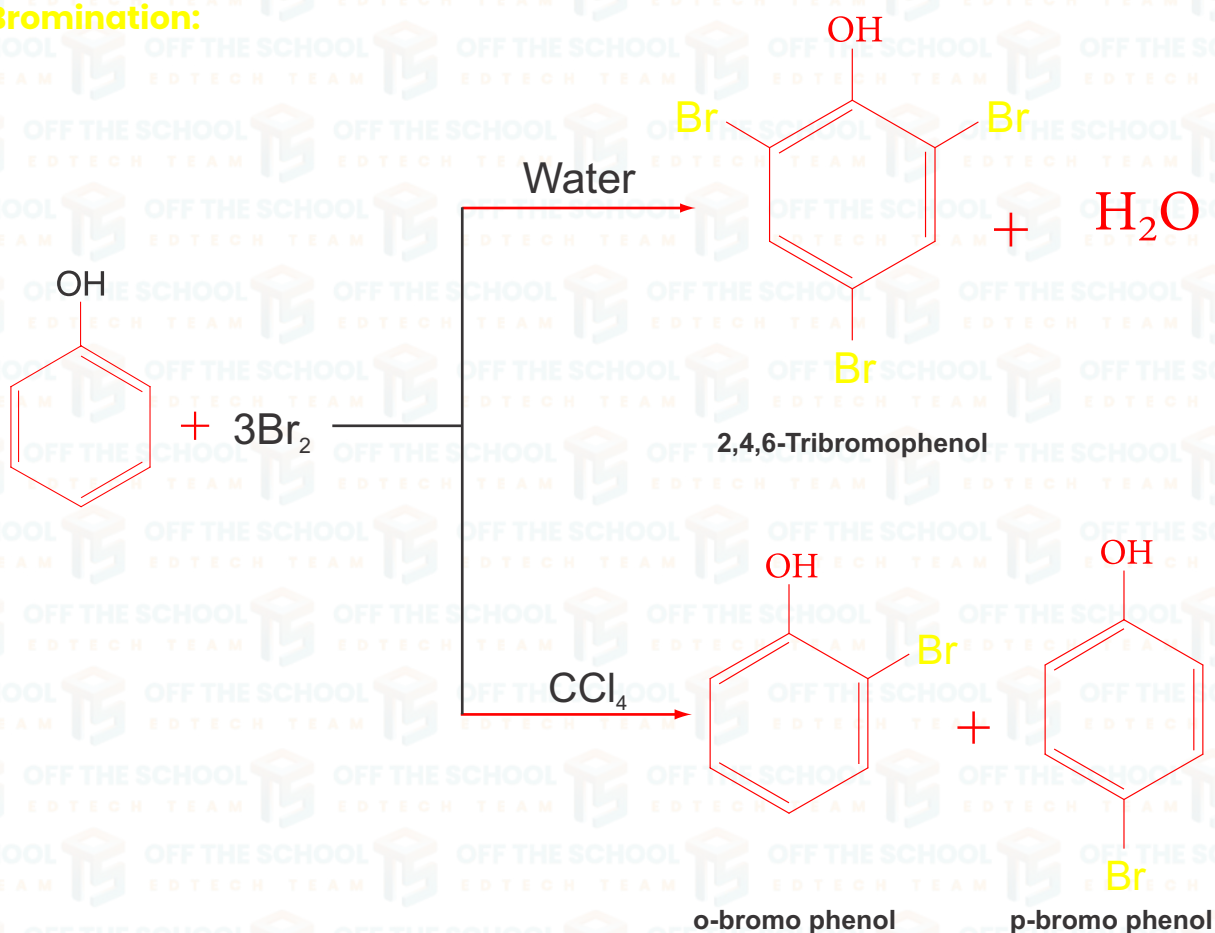
1. Nitration:



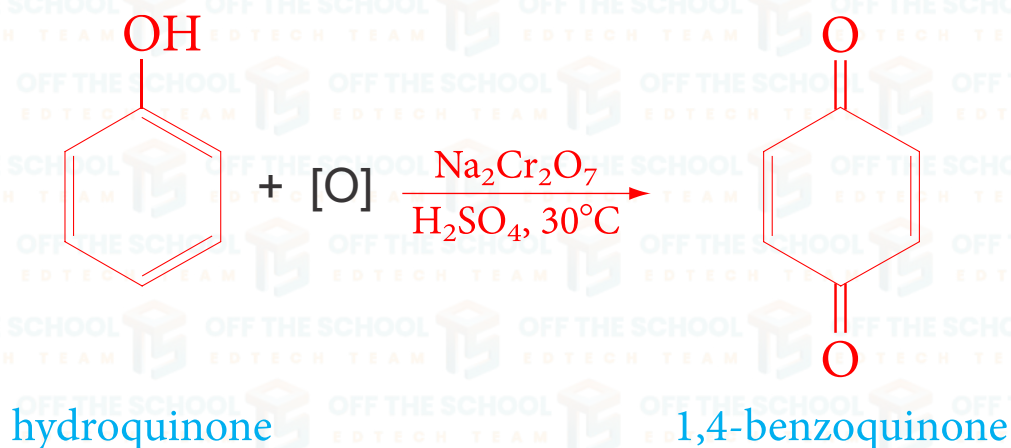
2. Sulphonation:



3. Bromination:



Oxidation in phenols:



Alcohols	Phenols
1. Generally less acidic.	More acidic due to the stability of phenoxide ion.
2. Lower boiling points for similar molecular sizes.	Higher boiling points due to additional aromatic ring.
3. Strong hydrogen bonding with water increases solubility.	Reduced solubility in water due to the aromatic ring.
4. The hydroxyl group ($-\text{OH}$) is attached to a saturated carbon atom.	The hydroxyl group is attached to an aromatic carbon atom.
5. Do not form colored complexes with iron (III) chloride.	Form colored complexes with iron (III) chloride, indicating presence.

CLASSIFICATION OF ETHERS:

On the basis of the nature of groups attached on either side of functional groups, ethers are classified as:

1. Simple or symmetrical ether: Two groups present on either side of the functional group are the same.

• eg: CH_3OCH_3 , $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$, $\text{C}_6\text{H}_5\text{OC}_6\text{H}_5$, etc.

2. Mixed or unsymmetrical ether: Two groups present on either side of the functional group are different.

• eg: $\text{CH}_3\text{OC}_2\text{H}_5$, $\text{C}_3\text{H}_7\text{OC}_6\text{H}_5$, etc.

Preparation of aliphatic ether:

1. Williamson's synthesis: Haloalkane is reacted with sodium or potassium alkoxide to give ether. This reaction is called Williamson's synthesis. It is used to prepare both symmetrical and unsymmetrical ether.

**2. Dehydration of Alcohol:****Oxonium Ion:**

Definition: An oxonium ion is a positively charged oxygen species (R_3O^+).

Formation from Ether: Ethers can form oxonium ions when they react with strong acids. The acid protonates the ether, resulting in the formation of the oxonium ion.

Chemical Equation: For example, diethyl ether forming an oxonium ion:

